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MMIEA: Multi-modal Interaction Entity Alignment model for knowledge graphs

Bin Zhu^{a,b}, Meng Wu^{c,d}, Yunpeng Hong^{c,d}, Yi Chen^{a,b}, Bo Xie^{a,b}, Fei Liu^{c,d}, Chenyang Bu^{c,d,*}, Weiping Ding^{e,*}^a State Key Laboratory of Pulsed Power Laser Technology, College of Electronic Engineering, National University of Defense Technology, Hefei, 230037, China^b Key Laboratory of Infrared and Low Temperature Plasma of Anhui Province, College of Electronic Engineering, National University of Defense Technology, Hefei, 230037, China^c Key Laboratory of Knowledge Engineering with Big Data (the Ministry of Education of China), Hefei University of Technology, Hefei, 230009, China^d School of Computer Science and Information Engineering, Hefei University of Technology, Hefei, 230009, China^e School of Information Science and Technology, Nantong University, Nantong, 226019, China

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ABSTRACT

Fusing data from different sources to improve decision making in smart cities has received increasing attention. Collected data through sensors usually exist in a multi-modal form, such as values, images, and texts. Thus, designing models that handle multi-modal data has an important role in this field. Meanwhile, security and privacy issues cannot be ignored, as the leakage of big data may provide opportunities for criminals. To solve the above challenges, we focus on research on multi-modal entity alignment for knowledge graphs and proposed the Multi-Modal Interaction Entity Alignment model (MMIEA). The model is proposed from the perspective of fusing data from different modalities while maintaining privacy. We determined that the model is privacy-preserving because it does not need to transmit the raw data of each modality (only the vector representation is transmitted). Specifically, we introduce and improve the BERT-INT model for the entity alignment task in multi-modal knowledge graphs. Experimental results on two commonly used multi-modal datasets show that our method outperforms 17 algorithms, including nine multi-modal entity alignment methods.

1. Introduction

The concept of a smart city refers to the collection and analysis of different types of data through various physical devices connected to the Internet of Things to optimize the city's operation and service efficiency [1]. Based on the integration, analysis, and processing of sensory data in all aspects of urban management, the development of smart cities is considered conducive to the promotion of scientific governance and rational layout of cities, and the convenience of urban life [2], such as smart transportation [3], smart campus [4], and smart emergency management [5].

An important aspect of smart city construction is the integration of data (such as remote sensing data) or knowledge from different sources to obtain a complete image of the city and to conduct data analysis and mining on this basis for scientific decision-making. Owing to the different types of sensors, the collected data usually exists in a multi-modal form, including images, texts, numbers, etc. Moreover,

even with the same sensor, different data can be generated for the same area owing to different observation angles or observation times (such as photos taken by satellites from different angles). Moreover, owing to the need to consider data privacy and security, sensor data from different units or companies cannot typically be shared directly, which causes additional difficulties. Therefore, fusing data or knowledge of different modalities from various sources, that is, the task of entity alignment (EA), while ensuring privacy is an important issue in the construction of smart cities.

EA is an important task in the field of knowledge graphs that aims to determine whether two objects (called entities) from different sources point to the same object in the real world. The entity alignment technique helps integrate data from multiple sources into a complete knowledge graph to obtain a large image of the city; thus, it is useful for many downstream tasks [6–10]. A knowledge graph refers to a

* Corresponding authors.

E-mail addresses: chenyangbu@hfut.edu.cn (C. Bu), dwp9988@163.com (W. Ding).<https://doi.org/10.1016/j.inffus.2023.101935>

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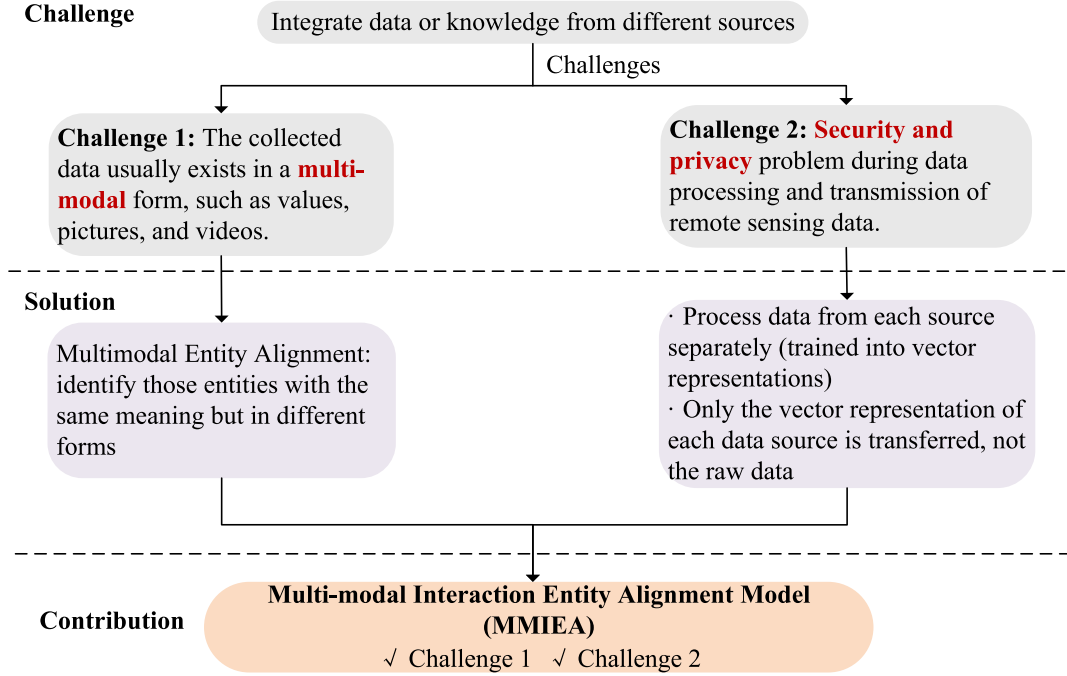


Fig. 1. Motivation and contribution of this study.

semantic network that describes objects (i.e., entities) and their relationships in the real world in the form of graphs [11]. Traditional knowledge graphs typically use only a single modality (i.e., text) to describe semantic information, whereas multi-modal knowledge graphs enrich the types of knowledge by introducing information from other modalities, such as images, into traditional knowledge graphs [12]. The construction of multi-modal knowledge graphs has been a hotspot in knowledge graph-related research in recent years [13].

EA models can be grouped into two categories: similarity-based methods and embedding-based methods [14–17]. Similarity-based methods are widely used in early research to judge whether two entities refer to the same object through textual similarity or propagation methods. Such methods have high accuracy, but the heterogeneity of knowledge graphs and complexity of multi-modal data have presented significant challenges. Embedding-based methods refer to training entities and relations into vector representations (called representation learning) and measuring the similarity between two entities according to the distance of the vectors. Embedding-based methods can capture the implicit structural semantics of the KG well. For example, the success of TransE [18] in the field of KG, particularly link prediction, EA, and other segment research fields, led to the development of a large number of related technologies. However, these methods do not pay sufficient attention to side information (data from different modalities), which hinders the accuracy of alignment.

To the best of our knowledge, most current research is on homogeneous data fusion (such as the alignment of different remote sensing images); only a small number of studies have focused on multi-modal entity alignment, for example, the fusion of images with other types of data [19]. Liu et al. [20] constructed a multi-modal knowledge graph MMKG by extracting or adding modal features, such as images, to the original datasets, and then proposed a multi-modal entity alignment framework called PoE. Liyi et al. [21] proposed a method MMEA to model the entity association of multi-modal KG, which consists of a multi-modal knowledge representation (MMKE) module and a multi-modal knowledge fusion (MMKF) module.

However, owing to the heterogeneity of knowledge graphs, equivalent entities often do not share similar neighbors. For example, in knowledge graph K_1 , Yun Ma may be associated with Alibaba Corporation and the founder of Alibaba, but in another knowledge graph K_2 , Jack Ma may be associated with Softbank Corporation and the director of Softbank [14]. Without sharing similar neighbors, it is difficult for the algorithm to correctly determine that “Yun Ma”, the founder of Alibaba, is the same person as “Jack Ma”, the former director of SoftBank Corp. For this case, where the corresponding neighbors cannot be found in K_1 , if a typical graph convolutional networks (GCN) based approach is adopted, the method of propagating different neighbors may introduce noise, thereby affecting the accuracy of the final alignment.

Inspired by BERT-INT [22], instead of aggregating neighbor nodes, we adopted the idea of computing the interactions between neighbor nodes to solve the above problem. Because the BERT-INT model does not specifically deal with multi-modal information, in this study, we improve the BERT-INT model by including a method for processing image modalities and image interactions, enabling it to handle multi-modal entity alignment tasks. The contributions of this study are as follows (see Fig. 1):

- Our work focuses on enhancing the BERT-INT model to address the entity alignment task in multi-modal knowledge graphs.
- We adopt an interactive approach to capture the image information of neighboring entities, leveraging this information for entity alignment. Moreover, we explore the impact of different fusion representations, namely concatenation and gating-based fusion methods, to achieve improved results in view fusion accuracy.
- Experimental results on two commonly used multi-modal datasets demonstrates that our method outperforms 17 algorithms, including 9 multi-modal EA methods.

The remainder of this paper is organized as follows. Section 2 summarizes related work. Section 3 reviews the preliminary work. In Section 4, we explain the proposed MMIEA model. Section 5 presents

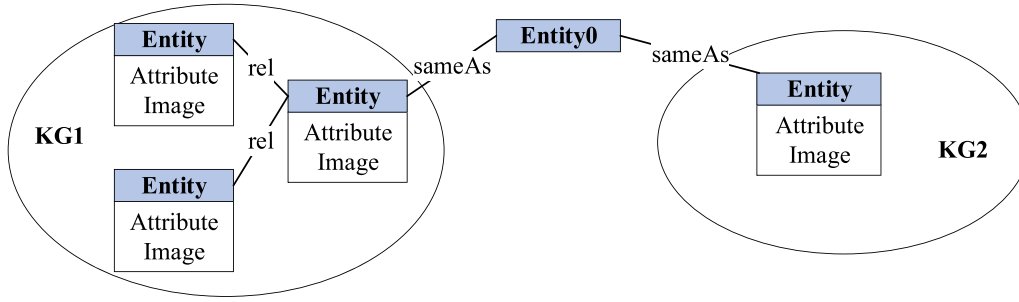


Fig. 2. Illustration of EA [15]. KG1 and KG2 are two different KGs. The goal of EA is to find all potential entity pairs (connected by “sameAs” relations) that can be aligned. For example, the entity “Kongzi” in Baidu Baike, the entity “Confucius” in Wikipedia and the entity “Kong Zhongni” in Hudong Baike all represent Confucius, the founder of Confucianism, in the real world.

the experimental results and analysis. Finally, Section 6 concludes the study.

2. Related work

Entity Alignment (EA) is a fundamental technology of KG, aiming to identify entities located in different KGs that refer to the same real-world object (see Fig. 2) [23–26]. In this section, we introduce the related work regarding EA for Single-modal and Multi-modal KGs.

2.1. Entity alignment for single-modal KGs

The approaches of EA for Single-modal KGs are introduced including the similarity-based, structure-based, and attribute-enhanced approaches.

Similarity-based Approaches. In the early years, EA was based on the idea of similarity: the more similar the text of an entity is, the more similar the entity is. For example, the editing-distance-based method uses word similarity, the TFIDF-based method [27] obtains word frequency similarity, and LDA [28] represents the category of the text model. Most of these methods are unsupervised and easy to use, and most are used in non-graph knowledge bases. This traditional similarity-based approach is becoming increasingly inappropriate as the structure of KG becomes complex and the amount of data increases.

Structure-based Approaches. In recent years, structure-based EA models have witnessed significant development, focusing on leveraging the structural information present in knowledge graphs (KGs). For instance, TransE [18], has led to the development of a large number of related technologies in KG research, particularly in the fields of link prediction and EA. MTransE [29] first learns the embedding representation of entities and relations within a single KG and subsequently learns a transfer matrix to map aligned entities from different KGs into a unified space. Similarly, IPTransE [30] initially learns the embedding of a single KG, and then iteratively refines the embedding through parameter sharing. Furthermore, in [31] a scalable GNN-based entity alignment approach is proposed to reduce KG size and alignment loss.

Attribute-enhanced Approaches. In addition to structural information, some studies have focused on improving EA using side information. Attribute-enhanced EA models have gained attention due to their ability to leverage additional edge information [23]. GCN-Align [32] is a novel method that combines side information, capturing both the structure and entity attributes of KG based on GCN encoding. Wu et al. [33] propose joint learning of entity and relation representations for EA, specifically considering heterogeneous KGs. BERT-INT [22] achieves remarkable performance in EA experiments by leveraging side information such as entity names, descriptions, and attributes through BERT-based pre-training. It surpasses many embedding models that solely rely on structural information. Additionally, Tam et al. propose a cross-lingual EA framework [34], which integrates various information types to maximize the utilization of cross-lingual KG data. Their framework employs a multi-order graph convolutional neural

network to capture entity relationship correlations, while attribute correlations are fused through a translation machine. Ge et al. introduce the EASY framework [35], which jointly fuses features derived from entity names and graph structural information to enhance EA accuracy. Zeng et al. [36] propose a collective EA framework based on reinforcement learning, utilizing representative features to capture different aspects of entity similarity in heterogeneous KGs. Sun et al. [37] propose a robust and adaptive embedding-based EA method, which improves the performance of traditional embedding methods.

Similarity-based EA are simple and easy to use, but they are limited in their ability to handle entities with different structures but the same semantic meaning. Structure-based EA methods, on the other hand, can address this issue by considering the structural characteristics of entities, but they often necessitate domain knowledge and strong reasoning capabilities. In contrast, attribute-enhanced EA techniques have gained considerable popularity in recent times due to their ability to enhance the accuracy of EA through the integration of both structural and contextual information.

2.2. EA for multi-modal KGs

For multi-modal KGs, researchers have gradually started paying attention to the design of EA. The design of entity alignment (EA) in multi-modal knowledge graphs (KGs) has garnered increasing attention among researchers. A crucial approach to tackle multi-modal EA is through the utilization of a multi-view paradigm [38].

MMEA [21] focuses on learning knowledge representations of relationships, visual images, and numerical images, followed by fusion of these representations. Liu et al. [39] proposed a method called Visual Pivoting, which leverages visual semantic representations for unsupervised EA. HMEA [13] embeds the structure and image of an entity into a hyperbolic space to reduce the distortion. MultiJAF [40] learns the embedding of structure, attributes, and images in the KG and incorporates a dedicated module to process numerical values. MCLEA [41] employs contrastive learning to facilitate the embedding of structure, attributes, and images in the KG. Li et al. [42] propose the use of embedding-based prompts for video and language pre-training for task-oriented dialogue systems. Lu et al. [43] present a robust embedding approach for multi-modal KG representation learning. Dost et al. [44] introduce a method for aligning and linking entity mentions in images, text, and knowledge bases. Ma et al. [45] develop UniTranSeR, a unified transformer semantic representation framework for multi-modal task-oriented dialogue systems. Additionally, Wang et al. [46] investigate the influence of visual context in multi-modal EA.

These studies highlight the importance of multi-modal EA as a critical research direction that carries substantial implications for enhancing EA accuracy and robustness. Overall, research on EA in multi-modal KGs is still at an early stage, and different methods exhibit specific limitations and challenges. Future research directions should delve deeper into exploring the effective utilization and integration of information from diverse modalities for EA.

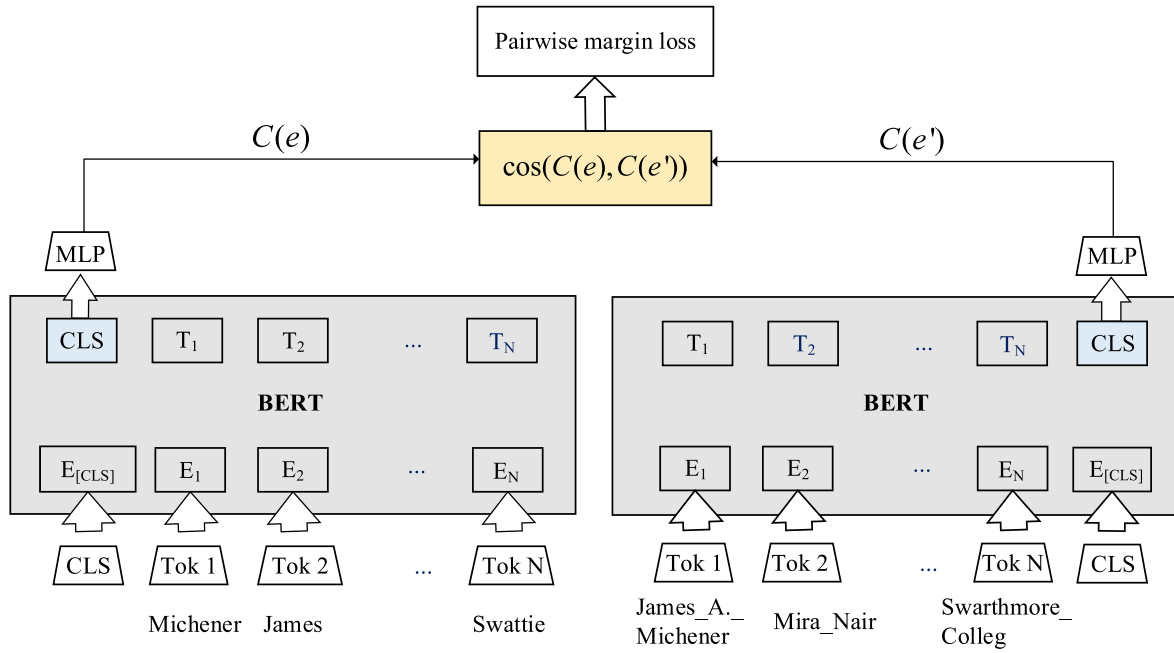


Fig. 3. Name/description-view in the BERT-INT model [22], including BERT unit. First, obtain the entities' vector representation with BERT and MLP. Then, calculate the cosine similarity as result of this view.

3. Preliminary

Tang et al. [22] proposed a BERT-based interaction model (BERT-INT) for single-modal KG. Before introducing our model, the details of the three views of BERT-INT are described. BERT-INT is a multiview model. It calculates the side information from the three views of the name/description (Section III-A), neighbor (Section III-B), and attribute (Section III-C), and then obtains the similarity between entities.

3.1. Name/description-view

The name/description view is based on the BERT unit. Entity alignment is the downstream objective for fine-tuning a pretrained BERT model. For each entity e in the dataset, a pre-trained BERT is applied to accept its name/description as the input. Then, we filter the classification (CLS) embedding of BERT by a multi-layer perceptron (MLP) layer to obtain the entities' vector representation $C(e)$. The calculation of $C(e)$ is given by Eq. (1) [22].

$$C(e) = MLP(CLS(e)), \quad (1)$$

where $CLS(e)$ returns the CLS embedding of the entity e . $MLP(x)$ returns the output of an MLP when feeding x .

Based on the BERT unit, the name/description view was first performed. This work is illustrated in Fig. 3. In this view, entities' descriptions are used as input (use entities' names when missing), and the vector representation of entity $C(e)$ is obtained after passing through the BERT unit. Then, the cosine similarity $\cos(C(e), C(e'))$ is calculated as the basis of EA.

3.2. Neighbor-view

In the neighbor view, we compare and calculate the neighbors $\mathcal{N}(e)$ of the entity e with the neighbor $\mathcal{N}(e')$ of the entity e' to obtain the similarity vector $\phi(\mathcal{N}(e), \mathcal{N}(e'))$ of the neighbors of the two entities. This method is an interactive method. It does not rely on aggregating neighbors' name/descriptions to learn a global representation for e or e' [33,47].

The calculation steps in the neighbor-view are as follows:

1. For each neighbor of entity e or e' , apply the BERT unit with name or description to obtain the collection of vector representation $\{C(e_i)\}_{i=1}^{|\mathcal{N}(e)|}$ and $\{C(e'_i)\}_{i=1}^{|\mathcal{N}(e')|}$.
2. Based on the collection of vector representation, calculate the neighboring entity similarity matrix S , where s_{ij} makes up S . The calculation of s_{ij} is presented in Eq. (2) [22].

$$s_{ij} = \frac{C(e_i) \cdot C(e'_j)}{\|C(e_i)\| \cdot \|C(e'_j)\|}. \quad (2)$$

3. For each relation of neighbors, apply the BERT unit with the relation's name to obtain $\{C(r_i)\}_{i=1}^{|\mathcal{N}(r)|}$ and $\{C(r'_i)\}_{i=1}^{|\mathcal{N}(r')|}$.
4. Based on Step 3, calculate the neighboring relation mask matrix M . The calculation of each $m_{ij} \in M$ is presented in Eq. (3) [22].

$$m_{ij} = \frac{C(r_i) \cdot C(r'_j)}{\|C(r_i)\| \cdot \|C(r'_j)\|}. \quad (3)$$

Understand the neighboring relation and neighboring entity as a pair of key values. For two pairs of key values pairs, if the value is very similar and the key is very similar, it can significantly increase the similarity of the two pairs of key values pairs.

5. Multiply M to S , i.e., $S_{ij} = S_{ij} \otimes M_{ij}$, where $\otimes$ indicates elementwise product.
6. S is processed using max-pooling and kernel for each row and column, respectively, to obtain the corresponding similarity vectors. These row similarity vectors and column similarity vectors are calculated separately, as shown in Eq. (4) [22].

$$\begin{aligned} s_i^{\max} &= \max_{j=0}^n \{s_{i0}, \dots, s_{ij}, \dots, s_{in}\}, \\ K_l(s_i^{\max}) &= \exp \left[-\frac{(s_i^{\max} - \mu_l)^2}{2\sigma_l^2} \right], \\ \mathbf{K}^r(S_i) &= [K_1(s_i^{\max}), \dots, K_l(s_i^{\max}), \dots, K_L(s_i^{\max})], \\ \phi^r(\mathcal{N}(e), \mathcal{N}(e')) &= \frac{1}{|\mathcal{N}(e)|} \sum_{i=1}^{|\mathcal{N}(e)|} \log \mathbf{K}^r c(S_i). \end{aligned} \quad (4)$$

7. Combine the two similarity vectors in the row and column. Obtain similarity vector ϕ for measuring the degree of matching the entities. The calculation of $\phi(\mathcal{N}(e), \mathcal{N}(e'))$ is presented in Eq. (5) [22].

$$\phi(\mathcal{N}(e), \mathcal{N}(e')) = \phi^r(\mathcal{N}(e), \mathcal{N}(e')) \oplus \phi^c(\mathcal{N}(e), \mathcal{N}(e')). \quad (5)$$

3.3. Attribute-view

The attribute view is similar to the neighbor-view. The only difference is the key-value pair. The name of the attribute is the key, and the content of the attribute is the value. The remainder is similar to the neighbor-view. It is worth noting that the attributes of the entity often have multiple pairs; therefore, we only need to consider several attributes of the current entity to interact without considering any neighbor information.

The calculation steps under the attribute view are as follows.

1. For each attribute of entity e or e' , apply the BERT unit with the attribute's value to obtain the collection of vector representation $\{C(a_i)\}_{i=1}^{|\mathcal{A}(e)|}$ and $\{C(a'_i)\}_{i=1}^{|\mathcal{A}(e')|}$.
2. Based on the collection of vector representation, calculate attributing entity similarity matrix S , where s_{ij} makes up S , $s_{ij} = \frac{C(a_i) \cdot C(a'_j)}{\|C(a_i)\| \cdot \|C(a'_j)\|}$.
3. For each attribute of entities, apply the BERT unit with the attribute's name to obtain $\{C(n_i)\}_{i=1}^{|\mathcal{A}(e)|}$ and $\{C(n'_i)\}_{i=1}^{|\mathcal{A}(e')|}$.
4. Based on Step 3, calculate attributing name mask matrix M , where m_{ij} makes up M , $m_{ij} = \frac{C(n_i) \cdot C(n'_j)}{\|C(n_i)\| \cdot \|C(n'_j)\|}$.
5. Multiply M with S , i.e., $S_{ij} = S_{ij} \otimes M_{ij}$.
6. S is processed using max-pooling and kernel for each row and column, respectively, to obtain the corresponding similarity vectors. These row similarity vectors and column similarity vectors are calculated separately. $\phi^r(\mathcal{A}(e), \mathcal{A}(e'))$ and $\phi^c(\mathcal{A}(e), \mathcal{A}(e'))$ are obtained, similar to the Eq. (4).
7. Combine the similarity vector in the row and column. Obtain the similarity vector $\phi(\mathcal{A}(e), \mathcal{A}(e'))$ for measuring the degree of matching the entities, similar to the Eq. (5).

4. MMIEA model

In this section, we introduce the MMIEA model in detail. This section contains our motivation and the detailed work on the image view. Fig. 4 illustrates the framework. The pseudocode for MMIEA is shown in Algorithm 1. By constructing four views, namely name/description, neighbor, attribute, and image, MMIEA achieves multi-modal entity alignment. In particular, MMIEA extends BERT-INT [22] to the multi-modal knowledge graph scenario by adding an image-view, providing richer information for entity alignment.

4.1. Motivation

Knowledge graph (KG) is a popular knowledge modeling technology in both industrial and academic areas. It is a data structure composed of nodes and edges to store knowledge data and has constructed knowledge bases or KGs covering various fields. Representative KGs include YAGO [48], FreeBase [49], DBpedia [50], and Omega [51]. They are widely used in popular fields such as search engines [6], question answering [7], recommendation systems [8,9], and natural language processing [10].

Most of the existing multi-modal entity alignment methods are embedding-based methods; that is, information of different modalities is projected into vectors first, and then the resulting vector representation is used for subsequent fusion or alignment. The method based

Algorithm 1 Framework of MMIEA

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1: Input: Entity  $e$  in  $KG$  and entity  $e'$  in  $KG'$ 
2: Output: Entity alignment result of  $e$  and  $e'$ 
3: if Name/Description information is present then
4:   Conduct work in Name/Description-View
5:   Gain  $\cos(C(e), C(e'))$ 
6: end if
7: if Neighbor-View information is present then
8:   Conduct work in Neighbor-View
9:   Gain  $\phi(\mathcal{N}(e), \mathcal{N}(e'))$ 
10: end if
11: if Attribute-View information is present then
12:   Conduct work in Attribute-View
13:   Gain  $\phi(\mathcal{A}(e), \mathcal{A}(e'))$ 
14: end if
15: if Image-View information is present then
16:   Conduct work in Image-View (Algorithm 2)
17:   Gain  $\phi(I(e), I(e'))$ 
18: end if
19: Conduct fusion process and gain entity alignment result

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on representation learning can effectively eliminate the heterogeneity between different modal information. Although the representations of different modalities may be in their own representation spaces, the structural consistency of the representation is the key to subsequent multi-modal information.

In multi-modal entity alignment based on embedding, a key issue is obtaining high-quality representations between various modalities. For traditional triple information, many high-quality models have been designed, such as the translation model TransE and its variants. However, current research has not explored the application of entity alignment to other modal information representation models, such as image information.

BERT-INT [22] is a highly effective EA model. It is based on the BERT [52] pretrained model to obtain entity embeddings and align them based on the similarity of the entity embeddings. Additionally, it uses an MLP for modal fusion, without the need to set weight hyperparameters.

Inspired by BERT-INT [22], we adopted the idea of computing the interactions between neighbor nodes to solve the problem that equivalent entities often do not share similar neighbors. Because the BERT-INT model does not specifically deal with multi-modal information, we improved the BERT-INT model, including a method for processing image modalities and image interactions, enabling it to handle multi-modal entity alignment tasks. Specifically, we use interactive techniques in image modality processing. In each interaction step, we calculate the image similarity matrix of the neighbor entities for two entities based on the current entity's neighbor set and neighbor image embeddings. Then, we process the rows and columns of the matrix using max-pooling and kernel functions to obtain the final image view vector.

4.2. Interaction of image-view

The interaction of the image views consists of two cases. If an entity has multiple images, we can refer to the attribute view for interaction. However, the current public dataset of multi-modal KG has only one image per entity, so we developed an alternative method of interaction. First, we borrow images of entity's neighbors as the basis and data for multigraph interactions. Then, the embedding vector of multiple images related to the entity constitutes the similarity matrix, similar to Eq. (2) and (3). Finally, use Eqs. (4) and (5) to aggregate and obtain the similarity under image-view. A diagram of the visual interaction is shown in Fig. 4(b). The detailed steps are explained in Section 4.3.

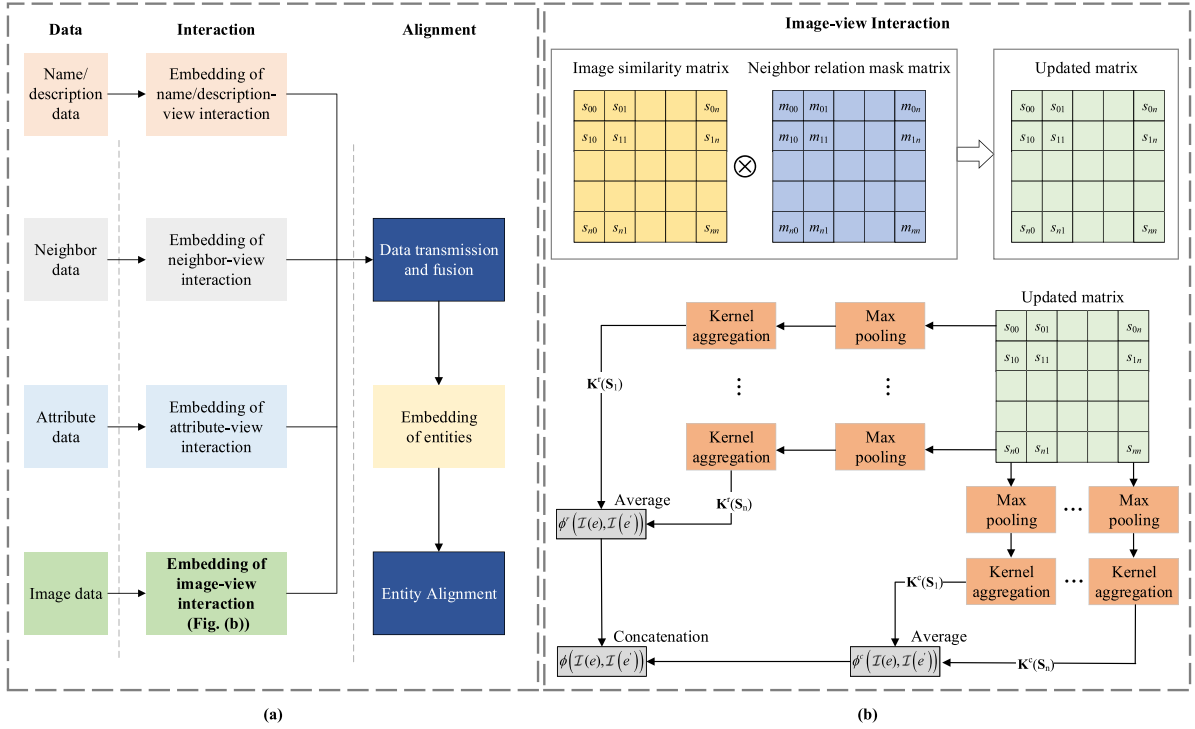


Fig. 4. Framework of our proposed MMIEA. (a) MMIEA introduces image views that contain rich information. The application of the framework in a remote sensing area can effectively protect the data privacy because the data of different modalities get embedded and transmitted separately. (b) Diagram of image-view. First, obtain the entities' vector representation with image model. Then, obtain the visual image similarity matrix S and neighboring relation mask matrix M . Finally, combine them and calculate similarity vector ϕ for measuring the degree of matching the entities under the image-view.

4.3. Image-view

We introduced an image modal to determine whether a pair of entities refers to the same real entity more accurately. For example, it is difficult to distinguish Apple Inc. from apple the fruit in the text “apple”, but it is easy to distinguish it from the images. The visual features are more intuitive and serve as the icing on the cake for EA. The image view is described in detail below:

First, we obtained the embedding vector of each image of the entities. We learn the embedding vectors of images according to the VGG16 [53] model. The filters in a stack of convolutional layers had receptive fields of 3×3 . There are 13 convolutional layers with different depths in the various architectures. The following are the three fully connected layers. Then, we obtain 4096-dimensional embedding vectors for all entities' images.

Given a pair $(e_i, i) \in Y$, where Y is the visual knowledge in KG, we use a score function to measure the visual features.

$$f_{img}(e_i, i) = -\|e_i - \tanh(\text{vec}(i))\|_2^2, \quad (6)$$

where $\text{vec}(i)$ denotes the projection and $\tanh(\cdot)$ denotes the activation function. Based on Eq. (6), we minimized the loss function as follows:

$$L_{img} = \sum_{(e_i, i) \in Y} \log(1 + \exp(-f_{img}(e_i, i))). \quad (7)$$

The pseudocode for the algorithm of the image view is shown in Algorithm 2.

The calculation steps under the image view are as follows.

- (1) For each entity e or e' , obtain the collection of embedding vectors $\{C(i_i)\}_{i=1}^{|I(e)|}$ and $\{C(i'_i)\}_{i=1}^{|I(e')|}$ of its neighboring images i_i , where the subscript i from 1 to $|I(e)|$ and $|I(e')|$ respectively.

Algorithm 2 Image-View

- 1: **Input:** Entity e in KG and entity e' in KG'
- 2: **Output:** Similarity vector ϕ of e and e'
- 3: **for** each neighboring image i of e and i' of e' **do**
- 4: calculate embedding vectors $C(i)$ and $C(i')$
- 5: **end for**
- 6: calculate visual image similarity matrix S
- 7: **for** each relation r and r' of e and e' **do**
- 8: calculate embedding vectors $C(r)$ and $C(r')$
- 9: **end for**
- 10: calculate neighboring relation mask matrix M
- 11: $S = S \otimes M$
- 12: Perform max-pooling and kernel on S , obtain ϕ^r and ϕ^c
- 13: Combine ϕ^r and ϕ^c to gain similarity vector ϕ

- (2) Based on the collection of embedding vectors, calculate visual image similarity matrix S , where s_{ij} makes up S , $s_{ij} = \frac{C(i_i) \cdot C(i'_j)}{\|C(i_i)\| \cdot \|C(i'_j)\|}$.
- (3) For each relation of neighbors, apply the BERT unit with the relation's name to obtain $\{C(r_i)\}_{i=1}^{|N(r)|}$ and $\{C(r'_i)\}_{i=1}^{|N(r')|}$.
- (4) Based on Step 3, calculate neighboring relation mask matrix M , where m_{ij} makes up M , $m_{ij} = \frac{C(r_i) \cdot C(r'_j)}{\|C(r_i)\| \cdot \|C(r'_j)\|}$.
- (5) Multiply M to S , i.e., $S_{ij} = S_{ij} \otimes M_{ij}$.
- (6) S is processed using max-pooling and kernel for each row and column, to obtain the corresponding similarity vectors. We obtain $\phi^r(I(e), I(e'))$ and $\phi^c(I(e), I(e'))$ according to the Eq. (5) in this step.

- (7) Combine the visual similarity vector in the row and column. Obtain the similarity vector $\phi(I(e), I(e'))$ for measuring the degree of matching the entities, similar to the Eq. (5).

4.4. Vector fusion

We use two approaches to fuse vectors, a direct concatenation operation and a gating-based fusion operation [54]. The following is an introduction to these two methods: concatenation fusion and gated fusion.

4.4.1. Concatenation fusion

we combine the work of the four views, the final fusion vector is expressed as follows:

$$\phi(e, e') = [\cos(C(e), C(e')) \oplus \phi(\mathcal{N}(e), \mathcal{N}(e')) \oplus \phi(A(e), A(e')) \oplus \phi(I(e), I(e'))] \quad (8)$$

4.4.2. Gated fusion

We use a gating-based fusion method to fuse the image view with other views. This model eliminates the need for manual parameter tuning and instead learns to adjust parameters directly from the training data. Here, x_v and x_t represent feature vectors associated with modalities v and t , respectively. Each feature vector is processed by a neuron (multiplicative gate) with a hyperbolic tangent (tanh) activation function to encode an internal representation feature h_v and h_t for the specific modality. For the input modalities x_v and x_t , a gate neuron employing the Sigmoid function controls the balance of the overall output derived from the computed features of x_v and x_t . The specific calculation formula is as follows:

$$\begin{aligned} h_v &= \tanh(W_v x_v) \\ h_t &= \tanh(W_t x_t) \\ z &= \sigma(W_z [x_v \oplus x_t]) \\ \phi(e, e') &= z * h_v + (1 - z) * h_t \end{aligned} \quad (9)$$

Among them, $x_v = \Phi(I(e), I(e'))$, $x_t = \cos(C(e), C(e')) \oplus \phi(\mathcal{N}(e), \mathcal{N}(e')) \oplus \phi(A(e), A(e'))$, $\phi(e, e')$ is the final fusion vector. the parameters in $\theta = \{W_v, W_t, W_z\}$ are learnable, and $\oplus$ represents the concatenation operation.

4.5. Loss function

We then obtain the final similarity scores of e and e' .

$$g(e, e') = \text{MLP}(\phi(e, e')). \quad (10)$$

The pairwise margin loss function is expressed as follows:

$$\mathcal{L} = \sum_{(e, e'^+, e'^-) \in D} \max\{0, g(e, e'^-) - g(e, e'^+) + m\}, \quad (11)$$

where the training data are $D = \{(e, e'^+, e'^-)\}$, and every triplet $(e, e'^+, e'^-) \in D$ contains a queried entity $e \in E$, rightly aligned counterpart $e'^+ \in E'$ and negative counterpart $e'^- \in E'$. m is the margin between the positive and negative pairs and $g(e, e')$ is instantiated as L_1 distance to measure the similarity between $C(e)$ and $C(e')$.

5. Experiments

To illustrate the validity of the proposed model and its modules, experiments were conducted to answer the following three main questions:

RQ1: Does our model outperform the state-of-the-art models in multi-modal EA applications?

RQ2: Does the multi-modal method and edge information in our model make sense in multi-modal EA applications?

RQ3: Does the fusion of multiple views in our model make sense in our model?

5.1. Setup

Datasets. We performed experiments on publicly available multi-modal datasets FB15k-DB15K, and FB15K-YAGO15k. Each dataset provides 20%, 50%, and 80% of the reference EA as the training sets. The basic statistics of the datasets are presented in Table 1.

Evaluation Metrics. Hits@n is the rate of correct entities ranked in the top n according to the similarity scores. We used Hits@1, Hits@5 and Hits@10 for our experiments. MRR denotes the mean reciprocal rank of the correct entity. Higher values of Hits@n and MRR explain the improved performance of the model.

Settings. The dimensions of BERT's CLS embedding were 768. We use a 300-dimension MLP in Eq. (1) and an 11 plus 1-dimension MLP in Eq. (10). The maximal numbers of neighbors and attributes in the KGs were both set to 50. If more, cut off; if less, add 0. The aggregation function was an RBF core aggregation function [55]. Image learning and embedding were consistent with MMEA.

Baselines. The baselines for the comparative experiment are as follows:

- **TransE** [18] is a typical KG embedding model.
- **MTransE** [29] learns the embedding representation of entities and relations in a single KG first, and then learns the transfer matrix to map the aligned entities from different KGs in the unified space.
- **IPTransE** [30] also learns the embedding of a single KG first, and then obtains the embedding of entities through iteration and parameter sharing.
- **SEA** [56] is a semi-supervised method, which uses labeled entities and abundant unlabeled entity information to align entities.
- **GCN-Align** [32] is a new way to combine edge information, which captures the structure and entity attributes of KG based on GCN coding.
- **IMUSE** [57] uses an unsupervised approach to generate a large number of high-quality aligned entities.
- **PoE** [20] combines multi-modal features and measures the reliability of matching by matching the underlying semantics of entities and mining the relations contained in the embedded space.
- **HMEA** [13] models structure and visual features in the hyperbolic space that can reduce distortion of learned multi-modal embeddings.
- **MMEA** [21] first learns the embeddings of multiple modalities separately and then maps these embeddings with separated spaces into a common space.
- **EVA** [58] jointly learns the multi-modal embeddings and combines these embeddings using the weight concatenation fusion method for final EA.
- **MCLEA** [41] is a multi-modal contrastive learning based EA model. It obtains effective joint representations for multi-modal EA and performs contrastive learning to jointly model intra-modal and inter-modal interactions.
- **MultiJAF** [40] model learns the embedding of structure, attributes, image in KG, and designs a special module to process the numerical value; progress has been made in the experiment of EA.
- **BERT-INT** [22] model is a BERT-based interaction model that performs entity alignment by computing interactions between neighboring entities.
- **DFMKE** [59] model uses a dual-fusion multi-modal knowledge graph embedding framework that combines early fusion and late fusion.
- **MEAformer** [60] is a multi-modal entity alignment transformer approach for meta modality hybrid, which dynamically predicts the mutual correlation coefficients among modalities for entity-level feature aggregation.

Table 1
Basic statistics of the datasets.

Datasets	Entities	Relations	Attributes	Relational triples	Attributes triples	Images
FB15K	14,951	1,345	116	592,213	29,395	13,444
DB15K	12,842	279	225	89,197	48,080	12,837
YAGO15K	15,404	32	7	122,886	23,532	11,194

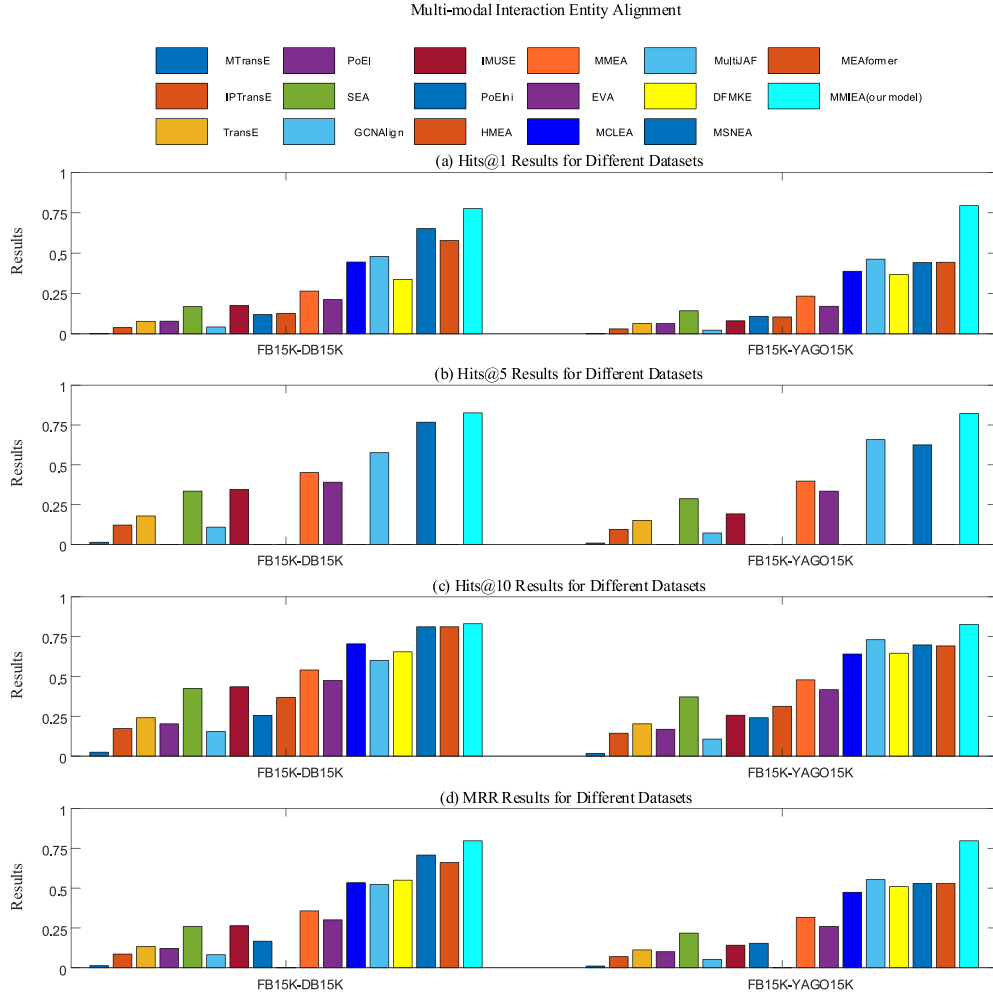


Fig. 5. Comparison of EA results of different models in the FB15K-DB15K and FB15K-YAGO15K datasets.

- **MSNEA** [61] designs a multi-modal knowledge embedding module to extract the visual, relational and attribute features of entities, making full use of multi-modal knowledge.

5.2. Performance analysis (RQ1)

We experimented with the latest multi-modal EA models. We divided these baselines into two categories: single-modal and multi-modal. The detailed experimental data are listed in Table 2. In Fig. 5, the abilities of the different models are clearly displayed in several pillar diagrams. The results of the comparison models were obtained from original studies to ensure that the results were realistic and valid.

Our model achieved the best results and was far superior to those of the other models. The reason for the optimal effect is that our model is multi-modal and uses a large amount of side information. Overall, the development of these baselines has progressed from single-modal to multi-modal, and from simple structure information to multi-modal information. Moreover, the accuracy of the models gradually improved. Experimental results show that our model is the best in the task of multi-modal EA, with a significant margin compared with state-of-the-art methods.

5.3. Comparison analysis (RQ2)

Multi-modal vs. Single-modal. In general, multi-modal models are significantly better than single-modal ones. The MMEA increased the metric of Hits@1 to 26.5%. Subsequently, MCLEA and MultiJAF increased Hits@1 of EA to 44.5% and 48.0%, respectively. The remaining metrics, including Hits@5, Hits@10, and MRR, showed remarkable improvements in MMEA, MCLEA, MultiJAF, and our MMIEA.

Side Information vs. Structural Information. The models with side information were better than those with only structural information. TransE, which considers only structural information, has a poor effect. However, embedded models such as MMEA, MCLEA and MultiJAF have made great progress because of the combination of more side information, particularly attributes, numerical values, and image information. Our MMIEA makes significant progress by using the BERT model to process side information. Therefore, it is better to use side information in EA.

Table 2
Results of comparison on multi-modal EA.

Type	Model	FB15K-DB15K				FB15K-YAGO15K			
		Hits@1	Hits@5	Hits@10	MRR	Hits@1	Hits@5	Hits@10	MRR
Single-modal	MTransE [29]	.003	.014	.025	.013	.003	.009	.017	.011
	IPTransE [30]	.039	.122	.173	.086	.031	.095	.144	.070
	TransE [18]	.078	.179	.241	.134	.064	.151	.203	.112
	PoE-I [20]	.079	–	.203	.122	.064	–	.169	.101
	SEA [56]	.169	.335	.425	.260	.141	.287	.372	.218
	GCN-Align [32]	.043	.109	.155	.082	.023	.072	.107	.053
	IMUSE [57]	.176	.346	.435	.264	.081	.192	.257	.142
Multi-modal	BERT-INT [58]	.775	.826	.832	.798	.793	.826	.830	.808
	PoE-Ini [20]	.120	–	.256	.167	.109	–	.241	.154
	HMEA [13]	.127	–	.369	–	.105	–	.313	–
	MMEA [21]	.265	.451	.541	.357	.234	.398	.479	.317
	EVA [58]	.213	.391	.475	.301	.171	.335	.417	.260
	MCLEA [41]	.445	–	.705	.534	.388	–	.641	.474
	MultiJAF [40]	.480	.576	.601	.523	.463	.658	.731	.554
	DFMKE [59]	.338	–	.654	.55	.367	–	.645	.51
	MSNEA [61]	.652	.768	.812	.708	.442	.625	.698	.529
	MEAformer [60]	.578	–	.812	.661	.444	–	.692	.529
	MMIEA (concatenation)	.776	.826	.831	.798	.793	.826	.831	.808
	MMIEA (gated)	.778	.827	.833	.799	.793	.827	.831	.808

Table 3
Ablation study on multi-modal EA with different component and training ratios.

Training ratios	Component	FB15K-DB15K				FB15K-YAGO15K			
		Hits@1	Hits@5	Hits@10	MRR	Hits@1	Hits@5	Hits@10	MRR
0.2	BERT unit	.6778	.7790	.8033	.7233	.6783	.7738	.7972	.7213
	name	.6780	.7797	.8040	.7237	.6783	.7738	.7972	.7212
	neighbor	.6008	.7218	.7564	.6570	.5345	.6679	.7091	.5973
	attribute	.7066	.8015	.8149	.7495	.7769	.8229	.8274	.7972
	image	.0633	.2142	.3503	.1555	.0808	.2161	.3213	.1631
	w/o image	.7757	.8266	.8321	.7981	.7937	.8263	.8306	.8082
	MMIEA (concatenation)	.7762	.8265	.8316	.7979	.7938	.8267	.8307	.8086
0.5	MMIEA (gated)	.7783	.8269	.8332	.7995	.7937	.8269	.8309	.8080
	BERT unit	.6993	.7918	.8069	.7404	.7346	.8110	.8303	.7701
	name	.6995	.7925	.8076	.7408	.7346	.8110	.8303	.7701
	neighbor	.6222	.7323	.7639	.6744	.5749	.7042	.7446	.6366
	attribute	.7179	.8089	.8192	.7592	.8144	.8535	.8573	.8321
	image	.0711	.2389	.3674	.1674	.0846	.2193	.3366	.1690
	w/o image	.7883	.8302	.8347	.8073	.8319	.8578	.8610	.8348
0.8	MMIEA (concatenation)	.7885	.8304	.8350	.8070	.8321	.8580	.8606	.8438
	MMIEA (gated)	.7797	.8325	.8352	.8082	.8331	.8594	.8607	.8449
	BERT unit	.7524	.8232	.8341	.7832	.7606	.8173	.8302	.7877
	name	.7524	.8232	.8341	.7832	.7606	.8173	.8302	.7877
	neighbor	.6644	.7645	.7960	.7121	.6068	.7107	.7507	.6589
	attribute	.7469	.8088	.8225	.7760	.8240	.8521	.8553	.8369
	image	.0731	.2417	.3752	.1722	.0920	.2362	.3497	.1808
0.8	w/o image	.8330	.8610	.8645	.8461	.8374	.8544	.8570	.8455
	MMIEA (concatenation)	.8318	.8614	.8657	.8456	.8383	.8552	.8575	.8459
	MMIEA (gated)	.8373	.8602	.8649	.8419	.8392	.8557	.8584	.8468

5.4. Multi-view analysis (RQ3)

We also performed a separate EA experiment on each view, called an ablation study, under different partitions of the training and test datasets. The ablation study results are presented in Table 3.

- **Validity of the BERT.** The BERT model performs reasonably well, and simply using BERT in experimentation can surpass the previous model. Our improvement over the previous model is largely due to the Bert model.
- **The Importance of Image.** Experiments using only image modes were not effective enough to support the whole EA tasks, possibly because of the low number and poor quality of images. But it is indisputable that the image modal has played a role as the icing on the cake.
- **The Image Data in the Dataset is of Poor Quality.** Although the addition of the image modality only resulted in a slight improvement of around 0.2%, it still showed some positive effect. When using the same dataset, MMEA achieved an improvement

of around 0.2% after incorporating the image modality, while HMEA only showed a 1% improvement after replacing image data through web crawling. Therefore, we believe that the performance of the image modality depends not only on the method itself but also on the quality of the data. Furthermore, the paper [62] also mentioned the poor performance of the image data in this dataset, as it introduced a significant amount of noise and even caused a decrease in model accuracy after incorporating the image modality.

5.5. Parameter analysis

In Fig. 6, we compare the impact of different numbers of the hidden nodes in the MLP layer for the fusion module. It can be found that our model is insensitive to this hyper-parameter and satisfactory results can be obtained for all cases.

In Table 4, we conducted ablation experiments with combinations of different views in the MMIEA model to verify the effectiveness of

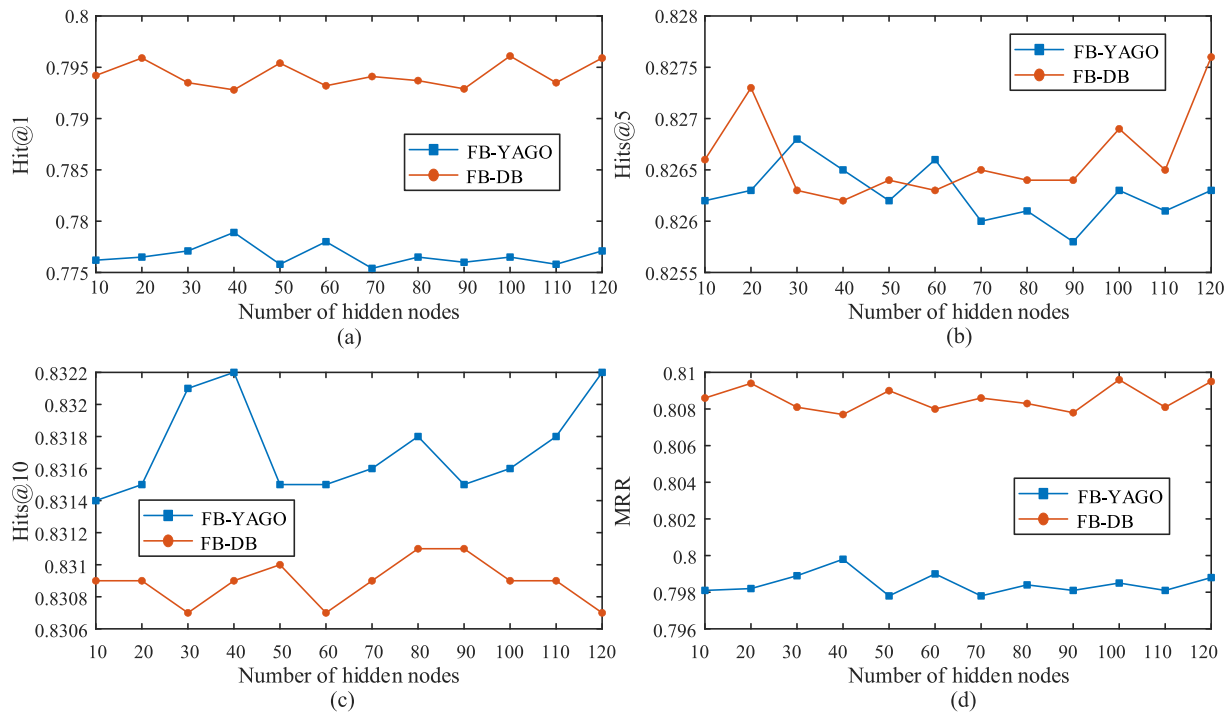


Fig. 6. Comparison results with different numbers of the hidden nodes for the fusion module.

Table 4
Ablation study of MMIEA with different combinations of views.

Component	FB-DB				FB-YAGO			
	Hits@1	Hits@5	Hits@10	MRR	Hits@1	Hits@5	Hits@10	MRR
name+image	.68166	.78320	.80695	.72724	.68054	.77743	.79942	.72340
neighbor+image	.59910	.72313	.75574	.65634	.53689	.67016	.71258	.59915
attribute+image	.71203	.80335	.81659	.75317	.77810	.82208	.82654	.79772

different views. The experimental results demonstrate that the model with the addition of picture views improves the accuracy, compared to a single view.

6. Conclusions

Entity alignment is fundamental for supporting the downstream application of the knowledge graph. Many researchers use the structural information of KG to align entities, but the side information, such as descriptions and attributes, is not sufficient. In particular, the use of multi-modal information such as images further affects the accuracy of entity alignment. To this end, we propose a multi-modal interaction entity alignment model (MMIEA) to completely utilize side information, including descriptions, attributes, and images. It calculates the degree of matching of entities in an interactive manner. Extensive experiments on two public datasets demonstrated the effectiveness of the MMIEA model with a significant margin compared to state-of-the-art methods.

CRedit authorship contribution statement

Bin Zhu: Conceptualization, Methodology, Review & editing. **Meng Wu:** Writing – original draft, Data curation, Visualization, Writing – review & editing. **Yunpeng Hong:** Data curation, Writing – review & editing. **Yi Chen:** Discussed the results, Writing – review & editing. **Bo Xie:** Discussed the results, Writing – review & editing. **Fei Liu:** Data curation, Writing – review & editing. **Chenyang Bu:** Conceptualization, Investigation, Supervision, Writing – review & editing. **Weiping Ding:** Methodology, Investigation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Bin Zhu Associate professor, deputy director of State Key Laboratory of Pulsed Power Laser Technology (Hefei China), National University of Defense Technology. He received his MS, PhD degrees in Optical Engineering and Signal and Information Processing from National University of Defense Technology, in 2007 and 2010 respectively. His study focuses on machine learning and image processing, including multi-modal information fusion, target and environment intelligent perception, etc.



Meng Wu received his bachelor's degree from Hefei University of Technology in 2020. He is currently working toward the master's degree with the School of Computer Science and Information Engineering, Hefei University of Technology, Hefei, China. His research interests lie in entity alignment and recommendation systems based on knowledge graphs.



Yunpeng Hong received the bachelor's degree from the Anhui University, HeFei, China, in 2022. He is currently working toward the masters degree with the School of Computer Science and Information Engineering, Hefei University of Technology, Hefei, China. His research interests include entity alignment and knowledge graph.



Yi Chen is a lecturer at the National University of Defense Technology. He received his MS and PhD degrees in Optical Engineering from the National University of Defense Technology in 2016 and 2020, respectively. His current research interests include optical imaging, computational imaging and computer vision.



Bo Xie is a lecturer at the State Key Laboratory of Pulsed Power Laser Technology, National University of Defense Technology, Hefei, China. He received his BS and MS degrees in Optical Engineering from the National University of Defense Technology in 2016 and 2018 respectively. His research interests include machine learning and image processing.



Fei Liu received her B.E. degree and M.E. degree in 2016 and 2019 from Hefei University of Technology, China, respectively. Now she is pursuing her Ph.D at the School of Computer Science and Information Engineering in Hefei University of Technology, China. Her research mainly lies in educational data mining and knowledge graphs. She has published articles in international conferences and journals, such as ACM KDD, IEEE Transactions on Fuzzy Systems (IEEE TFS) and IEEE Transactions on Emerging Topics in Computational Intelligence (IEEE TETCI). She is a PC member of WWW'22, AAAI'23.



Chenyang Bu is an associate professor at Key Laboratory of Knowledge Engineering with Big Data (the Ministry of Education of China), Hefei University of Technology, Hefei, China. He obtained the Ph. D degree from University of Science and Technology of China. He is a recipient of a Best Paper Award at IEEE International Conference on Knowledge Graph (ICKG) 2020 and a Best Student Paper Award at ICKG 2022. His research interests include knowledge graph construction and application, as well as automated graph learning with evolutionary algorithms.



Weiping Ding received the Ph.D. degree in Computer Science, Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 2013. From 2014 to 2015, he is a Postdoctoral Researcher at the Brain Research Center, National Chiao Tung University (NCTU), Hsinchu, Taiwan. In 2016, He was a Visiting Scholar at National University of Singapore (NUS), Singapore. From 2017 to 2018, he was a Visiting Professor at University of Technology Sydney (UTS), Ultimo, NSW, Australia. He is currently a professor with the School of Information Science and Technology, Nantong University, Nantong, China, and also the supervisor of Master and Ph.D postgraduate by the Faculty of Data Science at City University of Macau. His research interests include deep neural networks, multi-modal machine learning, granular data mining, and medical images analysis. In these areas, he has published over 200 scientific articles in refereed international journals, such as IEEE T-PS, T-NNLS, T-CYB, T-SMCS, T-BME, T-EVC, T-II, T-ETCI, T-CDS, T-ITS and T-AI. He has held 20 approved invention patents in total over 35 issued patents. He has co-authored two books. His fifteen authored/co-authored papers have been selected as ESI Highly Cited Papers (i.e. listed in Top 1% globally in the corresponding discipline). Dr. Ding is the Founding Chair of IEEE CIS Task Force on Granular Data Mining for Big Data. He is vigorously involved in editorial activities. He served/serves on the Editorial Advisory Board of Knowledge-Based Systems and Editorial Board of Information Fusion, Engineering Applications of Artificial Intelligence and Applied Soft Computing. He served/serves as an Associate Editor of IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Fuzzy Systems, IEEE/CAA Journal of Automatica Sinica, Information Sciences, Neurocomputing, Swarm and Evolutionary Computation, IEEE Access and Journal of Intelligent & Fuzzy Systems, and Co-Editor-in-Chief of Journal of Artificial Intelligence and System. He is the Leading Guest Editor of Special Issues in several prestigious journals, including IEEE Transactions on Evolutionary Computation, IEEE Transactions on Fuzzy Systems.